

There are three parts in our outline

- (1) Logic, logical and critical Reasoning
 - (2) Mathematical Modeling and analyses
 - (3) Statistical Modeling and analysis
-

(1) Logic, logical and critical reasoning

* Introduction and importance of logic

Logic is the study of reasoning. It's about understanding how to make arguments that are rational.

In simple terms, logic is like a set of rules for making sure that our thinking is clear, consistent, and leads to valid conclusions.

Example 1

* Premise 1 :- If students complete extra (assignmⁿ) assignments, they show initiative.

* Premise 2 : Ali completed extra assignments.

* Conclusion : Therefore, Ali showed initiative.

Example 2

Premise 1 : All apples are fruits.

Premise 2 : This is an apple.

Conclusion : This is a fruit.

* Inductive, deductive and adductive approaches of reasoning

Inductive reasoning

Inductive reasoning involves making decisions or drawing conclusions based on specific observations.

Example:- You noticed that every time it rains, the plant in the garden grows faster. Based on this observation, you conclude the rainwater is & beneficial for plant growth.

Deductive Reasoning

Deductive Reasoning involves drawing conclusion on the bases of premises by using logical rules.

Example:- Premise 1: All humans are mortal.

Premise 2: Socrates is human.

Conclusion: Therefore, Socrates is mortal.

Adductive Reasoning

Adductive Reasoning involves making educated guesses or hypotheses based on incomplete information.

Example:- You walk into your living room and notice that the TV is on, but there's no one else in the room. Then, you concluded that someone watched TV and did not switch off.

(1) Proposition:-

A proposition is a statement or assertion which can be either true or false.

(2) Truth value:-

A proposition must have a definite truth value either true or false.

Note:-

A proposition must be a statement, not a question, command or exclamation.

Examples:-

(i) The Earth revolves around the Sun.

(ii) John is taller than Sarah.

(iii) $5 > 9$

(iv) $13 > 2$

(3) Argument:- An argument is a ~~(serious)~~ series of statement or proposition in which one or more premises are used to support a conclusion.

Premise 1

All human are mortal.

Premise 2

Socrates is human.

Conclusion:-

Socrates is mortal.

(4)

(4) Logical connectives or logical operators

Connection	Symbol	Nature of the compound
And	$\wedge$	Conjunction
or	$\vee$	Disjunction
Not	$\sim$	Negation
Double Not	$\sim \sim$	Double Negation
iff		
If-then	$\rightarrow$ or $\Rightarrow$	Implication or Conditional
iff	$\leftrightarrow$ or $\Leftrightarrow$	Equivalence or biconditional

* Truth table of negation

p	$\sim p$	q	$\sim q$
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T

* Truth table of Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

* Truth table of Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

★ Truth table of double negation

P	$\sim P$	$\sim \sim P$
T	F	T
F	T	F

★ Truth table of Implication or Conditional

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

★ Truth table of biconditional

P	q	$P \iff q$
T	T	T
T	F	F
F	T	F
F	F	T

Logical Equivalence / Propositional Equivalence

statement

$$S_1(P, q, r, \dots), S_2(P, q, r, \dots)$$

$$1 \rightarrow \sim(P \wedge q) \equiv \sim P \vee \sim q$$

$$2 \rightarrow \sim(P \vee q) \equiv \sim P \wedge \sim q$$

$$3 \rightarrow \sim(P \Rightarrow q) \equiv P \wedge \sim q$$

$$4 \rightarrow \sim(P \Leftrightarrow q) \equiv (P \wedge \sim q) \vee (q \wedge \sim P)$$

Types of statement

1 Contingency statement

The statement which either true or false i.e whose true and false are mixup.

2 Tautology

The statement which is true in every case called tautology.

Ex: It will rain tomorrow or it will not rain.

③ Contradiction or absurdity

The statement which is false in every case called contradiction or absurdity.

It will rain tomorrow
and it will not rain.

$$\boxed{P \wedge \sim P}$$

P	$\sim P$	$P \wedge \sim P$
T	F	F
F	T	F

$\boxed{P \rightarrow q}$ Converse, Inverse and contrapositive

Converse : $q \rightarrow P$

Inverse : $\neg P \rightarrow \neg q$

Contrapositive: $\neg q \rightarrow \neg P$

Logical Fallacies

Logical fallacies are errors in reasoning that can make an argument or conclusion invalid or misleading.

There are some common types of logical fallacies:-

(i) Informal Fallacies

(1) Ad Hominem

Attack the person making the argument rather than addressing the argument itself.

Example: "You're just saying that because you're a Democrat / Republican."

(2) False Dichotomy

Presenting only two options when there are actually more.

Example: "You're either with us or against us".

(3) Slippery Slope

Arguing that a particular action will inevitably lead to a series of negative consequences without evidence.

Example: "If we allow same-sex marriage, next thing you know, people will be marrying animals."

(4) Appeal to authority

Using the opinion or position of an authority figure to support an argument without considering the validity of their claim.

Example:- "This celebrity endorses this product, so it must be good."

(ii) Formal Fallacies

(1) Affirming the consequent

~~Assuming~~ Assuming that if $A \Rightarrow B$ and B is true, then A must also be true.

Example:

Premise 1: If it rains, the ground gets wet ($A \Rightarrow B$)

Premise 2: The ground wet (B).

Conclusion: Therefore, it rained. (A)

(2) Denying the antecedent

Assuming that if $A \Rightarrow B$
and A is false, then B
must also be false.

Example:-

Premise 1: If it rains,
the ground gets wet. ($A \Rightarrow B$)

Premise 2: It did not rain ($\neg A$)

Conclusion: Therefore, the
ground is not wet. ($\neg B$)

Set

A collection of distinct objects is called a set.

Usually denoted by $A, B, C, \dots$.

Methods

There are three methods to represent the set.

(1) Tabular Method

The method in which we represent the set by its element as in table form.

(2) Descriptive method

The method in which we describe the set in words.

(3) Set builder Method

The method in which we describe the set in builder form called set builder method.

Operations in sets

① Union

② Intersection

③ Complement

④ Difference.

⑤ ① Subset: If each element of B is a member of set A then B is said to be a subset of A.

$$A = \{5, \dots, 100\}$$

$$B = \{20, \dots, 95\}$$

B is a subset of A.

$$B \subseteq A.$$

Super Set:- If end element of
of A and in this
case A is called superset of B.

$$A \supseteq B.$$

Note:- (i) Every set is a subset of itself.
(ii) Empty set is a subset of every set.
(iii) Every set has exactly two
improper subsets Φ and set itself.

Proper Subset

Set B is said to be
a proper subset of set A if
there exist at least one element
in A which is not in B.

Then B is a proper subset
of A. $B \subset A$.

Power Set of A:-

Set of all subsets of A.

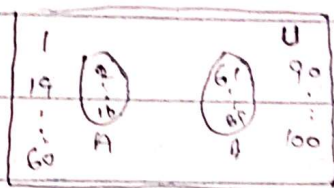
① Disjoint sets

The sets which have no common members.

$$U = \{1, \dots, 100\}$$

$$A = \{2, \dots, 100\}$$

$$B = \{61, \dots, 100\}$$



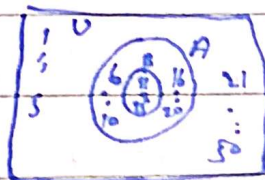
② Subsets:

$$B \subseteq A$$

$$U = \{1, \dots, 50\}$$

$$A = \{6, \dots, 20\}$$

$$B = \{11, \dots, 15\}$$

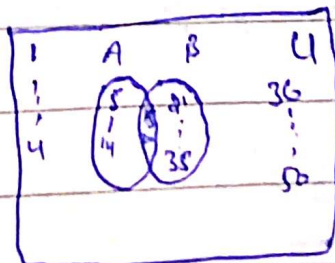


③ Overlapping sets -

$$U = \{1, \dots, 50\}$$

$$A = \{5, \dots, 20\}$$

$$B = \{15, \dots, 35\}$$



B is a subset of A.

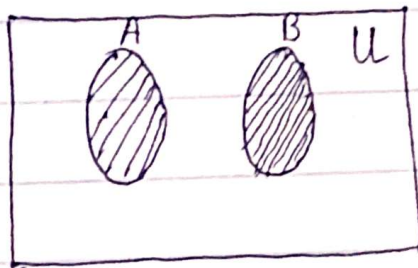
$B \subseteq A$

Shading

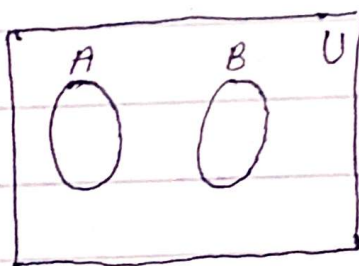
$A \cup B, A \cap B, A/B, B/A, A^c$

(1) Disjoint

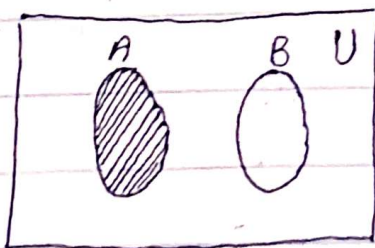
(i) $A \cup B$



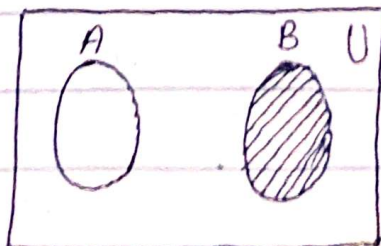
(ii) $A \cap B$



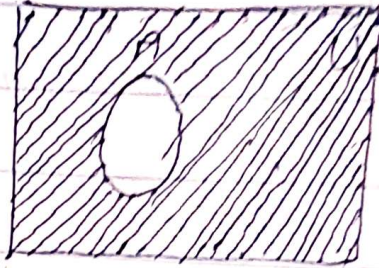
(iii) A/B



(iv) B/A

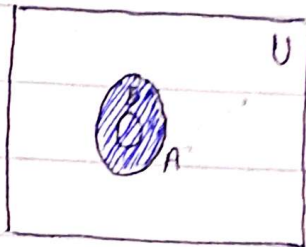


(v) A^c

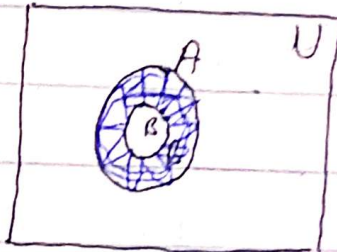


(2) Subset

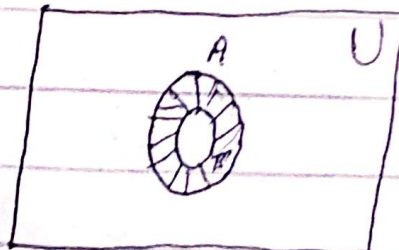
(i) $A \cup B$



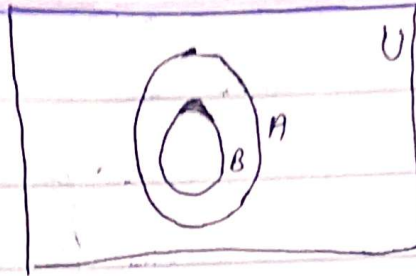
(ii) $A \cap B$



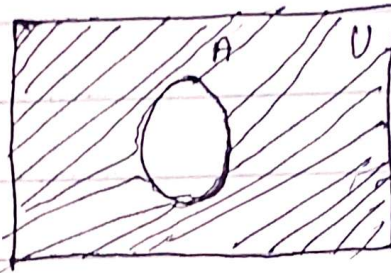
(iii) A/B



(iv) B/A

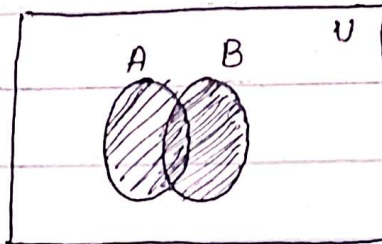


(v) A^c

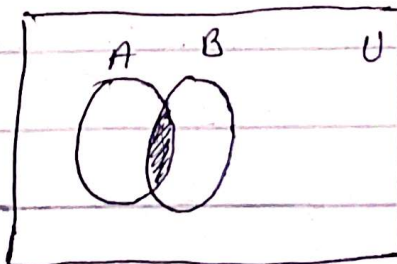


③ Overlapping

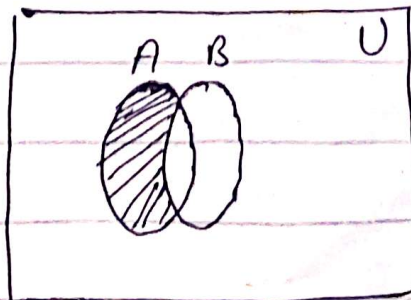
(i) $A \cup B$



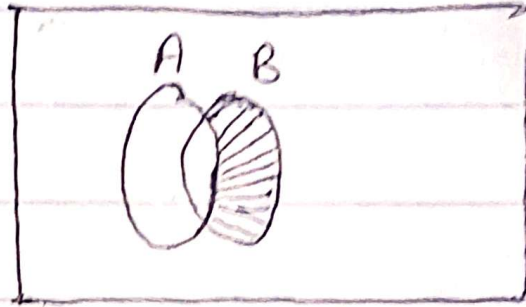
(ii) $A \cap B$



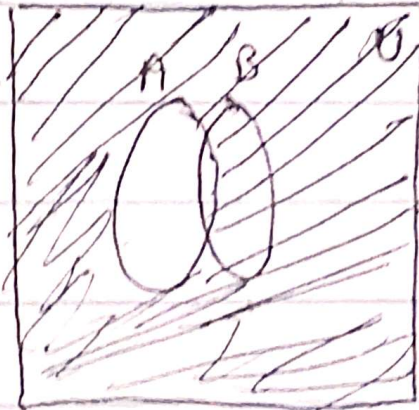
(iii) A/B



(iv) B/A



(v) A^c



* Predicates and quantifiers

Predicates

Predicates, predicts the statements.

For example:-

x is greater than y .

Predicate:-

Is greater than

① Quantification/Quantifier

(i) Existential Quantifier:-

$\exists$ there exist.

$\exists x, y$ such that $Q(x, y)$ is true.

(ii) Universal quantifier

$\forall$ for all

$P(x): x = x$

$\forall x, P(x)$ is true.

Examples:-

All human beings are mortal. let x denotes human beings.

$\forall x, P(x)$ is true.

② There exist x such that $x^2 = 5$

$\exists x, P(x)$ is true.

De Morgan's law

De Morgan's law is used in sets and logics.

The negation of conjunction is

$$(*) \quad \neg (P \wedge Q) = \neg P \vee \neg Q$$

equal to the disjunction of negations.

(*) The negation of disjunction is equal to the conjunction of the negations.

$$\neg (P \vee Q) = \neg P \wedge \neg Q$$

★ Quantitative reasoning exercises using logical reasoning concepts and techniques

Q 1 Determine with reasoning whether the given sentence is a proposition or a non-proposition

(1) The sun rises in the east.

Ans:- It is proposition.

(2) Close the door!

Ans:- It is a non-proposition due to a command and exclamatory sign.

(3) How are you today?

Ans:- It is non-proposition because it is a question.

(4) The Eiffel Tower is in London.

Ans:- It is a proposition, because it is not a question and command.

(5) Wow, this is amazing!

Ans:- It is not a proposition due to exclamation.

(6) $2 + 3 = 5$

Ans:- It is proposition.

Q2 Given the parameterized proposition $P(x)$: " x is a prime number", replace x with the values 2, 4, and 7. Indicate whether each resulting proposition is true or false.

Ans:-

Set of Prime numbers = $\{2, 3, 5, 7, 11, 13, \dots\}$

Now

$P(x)$: x is a prime number

Given numbers = 2, 4, 7

$P(2)$: 2 is a prime number True

$P(4)$: 4 is a prime number False

$P(7)$: 7 is a prime number True

Q3 Let $P(x)$: "The Capital of x is Washington, D.C." If x is "USA" and x is "Canada" determine the truth value of each proposition.

Ans:-

$P(x)$: "The capital of x is Washington D.C."

$P(\text{USA})$: "The capital of USA is / / / True

$P(\text{Canada})$: / / / Canada / / / False

Q4:- Let $P(x)$: " x is greater than 5" and $q(x)$: " x is an even number."

For $x = 4, 6, 8$ evaluate

$$P(x) \wedge q(x)$$

Ans:-

$P(x)$: " x is greater than 5"

$q(x)$: " x is an even number."

Given $x = 4, 6, 8$

For $x = 4$

$P(4)$: "~~4~~ 4 is greater than 5." False

$q(4)$: "4 is an even number." True

$$P(4) \wedge q(4) = \text{False}$$

For $x = 6$

$P(6)$: "6 is greater than 5". $\boxed{\text{True}}$

$Q(x)$: "6 is an even number". $\boxed{\text{True}}$

$$P(x) \wedge Q(x) = P(6) \wedge Q(6) = \text{True}$$

For $x = 8$

$P(8)$: "8 is greater than 5". $\boxed{\text{True}}$

$Q(8)$: "8 is an even number". $\boxed{\text{True}}$

$$P(8) \wedge Q(8) = \text{True} \wedge \text{True} = \boxed{\text{True}}$$

Q5 What does the proposition

$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x+y=0$ mean?

Ans:- This means 'for every real number x , there exist a real number y such that their sum is equal to zero. This is true because for any x , we can choose $y = -x$.'

Q6 Are the following propositions logically equivalent?

1. $\forall x \in \mathbb{N}, x^2 > x$

2. $\forall x \in \mathbb{N}, x > 1 \Rightarrow x^2 > x$

Ans:- The first statement is false because $(1)^2 \neq 1$ and second statement is true because as $x > 1$ the $x^2 > x$. So, these are not equivalent.

Q 7 Consider the proposition:
"For every continuous function $f(x)$, there exist a point $c \in [a, b]$ such that $f(c) = 0$."

Is this proposition always true?

Ans:- No, this is not always true. The proposition would only be true if $f(x)$ is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs. For example, $f(x) = x^2 + 1$ has no zero in $[a, b]$ for any real interval.

Q8 What is the truth value of the statement $P \wedge Q$ when P : "It is raining", Q : "The ground is wet."

Ans:- It is true.

P	Q	$P \wedge Q$
$\top$	$\top$	$\top$

Q9 A candidate qualifies for a senior position if P : "The candidate has at least 4 5 years of experience" and Q : "The candidate has a master's degree." Write the compound statement using $P \wedge Q$ and explain the conditions under which the candidate will qualify for the position.

Ans:-

The compound statement is "The candidate has at least 5 years experience and a master's degree."

$P \cap Q$

Q 10

A fair die is rolled.
Find the probability that the
outcome is both an even number
and greater than 3.

P: "The roll results is an
even number".

Q: "The roll results in a
number greater than 3."

Ans:-

Outcomes of fair die = $\{1, 2, 3, 4, 5, 6\}$

P : 2, 4, 6

Q : 4, 5, 6

	P	Q	$P \cap Q$
Outcomes			
1	F	F	F
2	T	F	F
3	F	F	F
4	T	T	T
5	F	T	F
6	T	T	T

Q 10 Consider the following propositions related to the stock market:

P: "The product is available in the store."

Q: "The product is available for the delivery."

Ans:-

P: "The product is available in the store." $\boxed{\text{True}}$

Q: "The product is available for the delivery." $\boxed{\text{True}}$

$$\underline{P \vee Q = \text{True}}$$

Q 12 If $A = \text{"It is raining"}$ and $B = \text{"It is cold"}$, translate the following statement into logical terms and simplify using De Morgan's law:

"It is not true that it is both raining and cold."

Ans:

A = "It is raining."

B = "It is cold."

According to given condition

$$\neg(A \wedge B) = \neg A \vee \neg B$$

Q13 Use De Morgan's Laws to simplify the expression $\neg(\neg A \vee B)$.

Ans:—

$$\begin{aligned}\neg(\neg A \vee B) &= \neg \neg A \wedge \neg B \\ &= \boxed{A \wedge \neg B}\end{aligned}$$

Q14 Simplify the following complex logical expression using De Morgan's Law: $\neg[(A \vee B) \wedge (C \vee D)]$

$$\begin{aligned}\neg[(A \vee B) \wedge (C \vee D)] &= \neg(A \vee B) \vee \neg(C \vee D) \\ &= (\neg A \wedge \neg B) \vee (\neg C \wedge \neg D)\end{aligned}$$

Q 15 Prove the equivalence
 $\neg(\neg A \vee \neg B) = A \wedge B$

$$\begin{aligned} \text{L.H.S} &= \neg(\neg A \vee \neg B) \\ &= \neg \neg A \wedge \neg \neg B \\ &= \boxed{A \wedge B} = \text{R.H.S} \end{aligned}$$

Hence proved

$$\underline{\text{L.H.S} = \text{R.H.S}}$$

Q 16 "All roses are flowers."
What is the observation of
this statement?

What is the conversion of
this statement?

Ans:-

Observation: No roses are non-flowers
Conversion: Some flowers are roses.

Q 17 "No dogs are cats."

What is observation?

What is contrapositive of this
statement?

Ans:-

Observation:- All dogs are not cats.
Contrapositive:- No cats are dogs.

Q 18 "Some books are novels."

- (1) What is conversion of this statement?
- (2) Can this be negated to say, "No books are novels"?

Ans:-

- (1) Conversion: Some novels are books.
- (2) No, this is logically in-valid.

Q 19 (1) "All apples are fruits"
(2) "Some fruits are sweets"

- (i) Are some apples sweet?
- (ii) Are all sweet things fruits?

Ans:-

- (i) Cannot be calculated
- (ii) False



Q 20 (i) "All squares are rectangles"

(ii) "Some rectangles are blue."

(i) Are some squares blue?

(ii) Are all rectangles squares?

Ans:-

(i) Cannot be calculated

(ii) False

Q 21 Statement: "Should all schools have a uniform dress code?"

Arguments:-

Ans:-

Arguments

(i) Yes a uniform ^{dress} code promotes equality and freedom.

(ii) No, students should have the freedom to choose their own clothes.

• 1st argument is strong. Ans

Q22 "Should online education be made compulsory for all students?"

Arguments = ?

Ans :-

(i) Yes, online education is more flexible and accessible to students across different regions.

(ii) No, online education lacks the personal interaction and practical experience that traditional classrooms provides.

Both arguments are strong.

Q23 In a class there are
• 8 students who play football and cricket

• 7 students who play football or cricket

• 13 students who play cricket

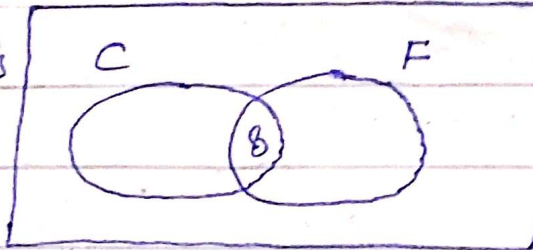
• 19 students who play football.

How many students are there in the class?

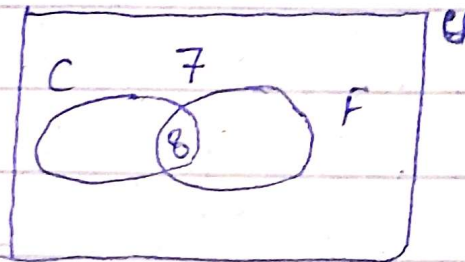
Ans:-

We can solve this
by using Venn diagrams

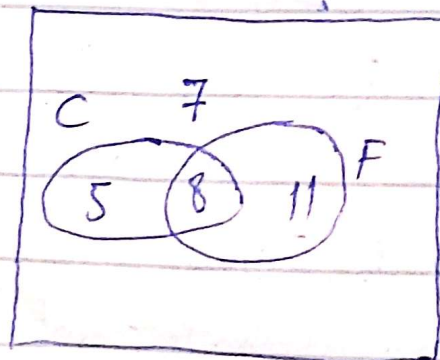
As 8 students
play both
cricket and
football.



As 7 students
play cricket or
football so



As 13 students
play only cricket
so we minus
the students who
play both.



$$13 - 8 = 5$$

As 19 students
play football only so
we minus 8 from 19.

$$19 - 8 = 11. \quad \text{Now}$$

Total students

$$\text{Total} = 7 + 5 + 8 + 11 = \boxed{31}$$

Q24 In a group of six friends everybody plays football or basketball. 4 members of the group play both sports and 1 member of the group plays only basketball. How many play only football?

Ans:-

Total number of friends = 6

How many plays basketball

and football = 4

Member play basketball = 1

Total Member play basketball = 4 + 1
= 5

Now, using the principle of inclusion-exclusion

$$|F \cup B| = |F| + |B| - |F \cap B|$$

$$6 = |F| + 5 - 4$$

$$6 = |F| + 1$$

$$6 - 1 = |F|$$

$$|F| = 5$$

These members play football
and basket - ball ~~both~~.

Now

Members play football = $5 - 4$

= $\boxed{1}$ Ans
